

~~2- Given~~

$$m = 2 \text{ Kg}$$

- a) $\vec{F}_1 = 3\hat{i} + 4\hat{j}$
 b) $\vec{F}_2 = -3\hat{i} + 4\hat{j}$
 c) $\vec{F}_2 = 3\hat{i} - 4\hat{j}$

$$a = ?$$

Sol.

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 \quad (\text{Newton's 2nd law})$$

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\vec{a} = \frac{\vec{F}_{\text{net}}}{m}$$

$$b) \quad \vec{F}_1 + \vec{F}_2 = [3\hat{i} + 4\hat{j}] + [-3\hat{i} + 4\hat{j}]$$

$$\vec{F}_{\text{net}} = (8\text{ N})\hat{j}$$

$$\vec{a} = 8/2$$

$$\boxed{\vec{a} = 4 \text{ m s}^{-2}}$$

$$a) \quad \vec{F}_1 + \vec{F}_2 = [3\hat{i} + 4\hat{j}] + [-3\hat{i} - 4\hat{j}]$$

$$= 0$$

$$c) \quad \vec{F}_1 + \vec{F}_2 = [3\hat{i} + 4\hat{j}] + [3\hat{i} - 4\hat{j}]$$

$$= (6\text{ N})\hat{i}$$

$$\vec{a} = 6/2 \Rightarrow \boxed{3 \text{ m s}^{-2}}$$

3- Given

$$m = 1 \text{ Kg}$$

$$a = 2 \text{ ms}^{-2}$$

$$\theta = 20^\circ$$

- a) x-component of net force
- b) y-component of net force
- c) net force in unit vector notation

$$\vec{F}_{\text{net}} = m\vec{a}$$

Sol

$$\begin{aligned} \text{a) } F_x &= ma_x \\ &= m a \cos \theta \\ &= (1)(2 \cos 20^\circ) \end{aligned}$$

$$F_x = 1.88 \text{ N}$$

$$\begin{aligned} \text{b) } F_y &= ma_y \\ &= (1)(2 \sin 20^\circ) \end{aligned}$$

$$F_y = 0.684 \text{ N}$$

$$\begin{aligned} \text{c) } \vec{F} &= F_x \hat{i} + F_y \hat{j} \\ \vec{F} &= 1.88 \hat{i} + 0.684 \hat{j} \end{aligned}$$

4- Given

$$\vec{F} = 3\hat{i} - 4\hat{j}$$

$$\vec{F}_1 = 2\hat{i} - 6\hat{j}$$

$$\vec{F}_2 = ?$$

because velocity is constant so $a=0$

$$F_{\text{net}} = ma$$

$$\vec{F}_1 + \vec{F}_2 = m(0)$$

$$\vec{F}_1 = -\vec{F}_2$$

$$\vec{F}_2 = -\vec{F}_1$$

Sol

$$\boxed{\vec{F}_2 = -2\hat{i} + 6\hat{j}}$$

5- Given

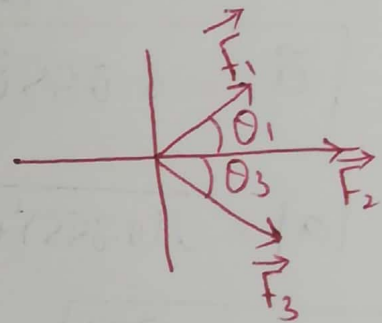
$$F_1 = 32\text{ N}, m = 120\text{ kg}$$

$$F_2 = 55\text{ N}$$

$$F_3 = 41\text{ N}$$

$$\theta_1 = 30^\circ$$

$$\theta_3 = 60^\circ$$



a) $\vec{a}$ in unit vector notation

b) Magnitude of $\vec{a}$

c) A direction relative to +ve x-axis.

d) Sol.

$$\vec{F}_1 = F_{1x} + F_{1y}$$

$$= 32(\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j})$$

$$\vec{F}_1 = 27.7\hat{i} + 16\hat{j}$$

$$\begin{aligned}\vec{F}_2 &= \vec{F}_{2x} + \vec{F}_{2y} \\ &= 55 (\cos 0^\circ \hat{i} + \sin 0^\circ \hat{j}) \\ \vec{F}_2 &= 55 \hat{i}\end{aligned}$$

$$\begin{aligned}\vec{F}_3 &= \vec{F}_{3x} + \vec{F}_{3y} \\ &= 41 (\cos(-60^\circ) \hat{i} + \sin(-60^\circ) \hat{j})\end{aligned}$$

$$F_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = m\vec{a}$$

$$\begin{aligned}\vec{a} &= \frac{\vec{F}_1 + \vec{F}_2 + \vec{F}_3}{m} \\ &= \frac{102.7 \hat{i} - 19.5 \hat{j}}{120}\end{aligned}$$

$$\boxed{\vec{a} = 0.855 \hat{i} - 0.1625 \hat{j}}$$

$$b) |\vec{a}| = \sqrt{(0.855)^2 + (-0.1625)^2}$$

$$\boxed{a = 0.881}$$

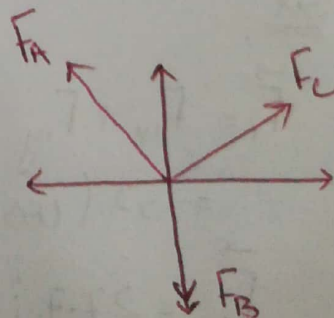
$$c) \theta = \tan^{-1}\left(\frac{-0.1625}{0.855}\right)$$

$$\boxed{\theta = -11^\circ}$$

b-

$$F_{\text{net}} = \vec{F}_A + \vec{F}_B + \vec{F}_C = m\vec{a}$$

$$F_{\text{net}} = 0$$



$$F_{\text{net},x} = F_c \cos 47^\circ - F_A \cos 47^\circ = 0$$

$$F_{\text{net},y} = F_c \sin 47^\circ + F_A \sin 47^\circ - F_B = 0$$

$$F_c \cos 47^\circ - F_A \cos 47^\circ = 0$$

$$F_c \cos 47^\circ = F_A \cos 47^\circ$$

$$47^\circ = \cos^{-1} \left(\frac{F_A \cos 47^\circ}{F_c} \right)$$

$$\boxed{47^\circ = 28.0^\circ}$$

Now,

$$F_c \sin 47^\circ + F_A \sin 47^\circ - F_B = 0$$

$$\boxed{F_B = 241 \text{ N}}$$

where

$$F_A = 220 \text{ N}$$

$$\theta = 47^\circ$$

$$F_{Ax} = 220 \cos 47^\circ \Rightarrow \boxed{150^\circ}$$

$$F_{Ay} = 220 \sin 47^\circ \Rightarrow \boxed{}$$

7- Given

$$\vec{F}_1 = 20 \text{ N}$$

$$a = 12 \text{ m/s}^2$$

$$\theta = 30^\circ$$

$$m = 2 \text{ Kg}$$

$$\vec{F}_2 = ?$$

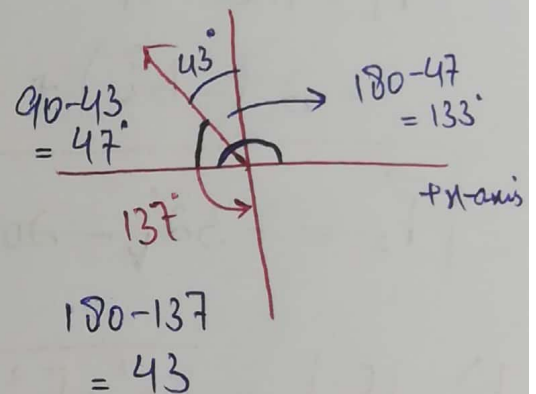
$$\therefore 180 - 137 = 43$$

$$90 - 43 = 47$$

$\nearrow$
x-axis

$$180 - 47 = 133$$

$\nearrow$
x-axis



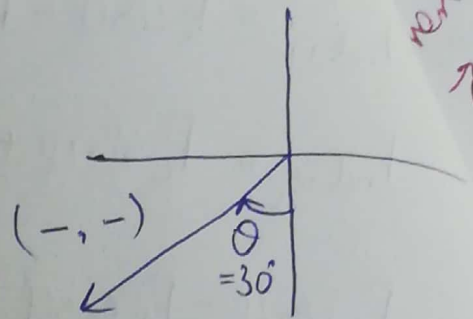
Sol

$$\vec{F}_1 + \vec{F}_2 = m\vec{a}$$

$$\vec{F}_2 = m\vec{a} - \vec{F}_1$$

a)

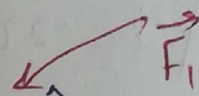
$$\vec{a} = -(12 \sin 30^\circ)\hat{i} - (12 \cos 30^\circ)\hat{j} \Rightarrow -a_x\hat{i} - a_y\hat{j}$$
$$\boxed{\vec{a} = -6\hat{i} - 10.4\hat{j}}$$



$$\vec{F}_2 = m\vec{a} - \vec{F}_1$$

$$= 2(-6\hat{i}) + 2(-10.4\hat{j}) - 20\hat{i}$$

$$\boxed{\vec{F}_2 = -32\hat{i} - 20.8\hat{j}}$$

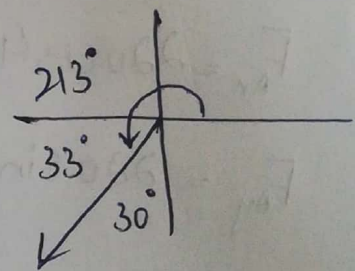


$$b) |\vec{F}_2| = \sqrt{F_{2x}^2 + F_{2y}^2}$$
$$= \sqrt{(-32)^2 + (-20.8)^2}$$
$$|\vec{F}_2| = 38.2 \text{ N}$$

$$c) \theta = \tan^{-1}\left(\frac{-20.8}{-32}\right)$$

$$\theta = 33.0$$

$$33^\circ + 180^\circ = 213^\circ \text{ (counter clockwise)}$$



men

$$\vec{a} = -8\hat{i} + 6\hat{j}$$

$$m = 2 \text{ Kg}$$

$$F_1 = 30\hat{i} + 16\hat{j}$$

$$F_2 = -12\hat{i} + 8\hat{j}$$

$$F_3 = ?$$

Sol

$$F_1 + F_2 + F_3 = m\vec{a}$$

$$F_3 = m\vec{a} - F_1 - F_2$$

$$= [2(-8\hat{i}) + 2(6\hat{j})] - 30\hat{i} - 16\hat{j} + 12\hat{i} - 8\hat{j}$$

$$\boxed{\vec{F}_3 = -34\hat{i} - 12\hat{j}}$$

9- Given

$$m = 0.34 \text{ Kg}$$

$$x(t) = -15 + 2t - 4t^3$$

$$y(t) = 25 + 7t - 9t^2$$

At $t = 0.7 \text{ sec}$

- magnitude of net force
- Angle from +ve x-axis of net force.
- Angle's of particle's direction of travel

Sol

$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{r} = x(t)\hat{i} + y(t)\hat{j}$$

$$\vec{v} = \frac{d}{dt}(-15 + 2t - 4t^3)\hat{i} + \frac{d}{dt}(25 + 7t - 9t^2)\hat{j}$$

$$\vec{v} = (2 - 12t^2)\hat{i} + (7 - 18t)\hat{j}$$

$$\vec{a} = \frac{d}{dt}(2 - 12t^2)\hat{i} + \frac{d}{dt}(7 - 18t)\hat{j}$$

$$= (-24t)\hat{i} + (-18)\hat{j}$$

at $t = 0.7$

$$\vec{a} = (-16.8)\hat{i} + (-18)\hat{j}$$

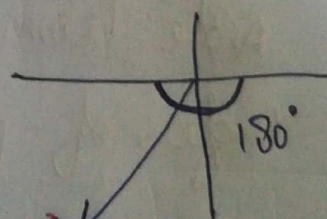
$$\vec{F} = m\vec{a} \Rightarrow [-5.712\hat{i} - 6.12\hat{j}]$$

$$|\vec{F}| = \sqrt{(-5.712)^2 + (-6.12)^2} \Rightarrow [8.37 \text{ N}]$$

$$b) \theta = 47^\circ \quad (\theta = \tan^{-1}(\frac{F_y}{F_x}))$$

where $\theta = 47^\circ - 180^\circ \Rightarrow -133^\circ$

(This shows that $\vec{F}$ is in 3rd Quadrant) $\vec{F}$

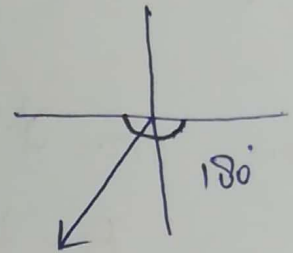


The direction of travel is the direction of tangent to path, which is the direction of velocity

$$\vec{r}(t) = (2 - 12t^2)\hat{i} + (7 - 18t)\hat{j} \quad \text{at } t = 0.7$$

$$\rightarrow \vec{v}(t) = -3.88\hat{i} - 5.60\hat{j}$$

$$\theta_v = \tan^{-1}\left(\frac{-5.60}{-3.88}\right) = 55.3^\circ$$



$$\theta_v = 55.3^\circ - 180^\circ$$

$$|\theta_v = -125^\circ| \quad \text{in } 3^{\text{rd}} \text{ Quadrant}$$

10- Given

$$m = 0.150 \text{ Kg}$$

$$x(t) = -13 + 2t + 4t^2 - 3t^3$$

$$\vec{F}_{\text{net}} = ?$$

$$\text{At } t = 3.40$$

Sol

$$\vec{v}(t) = \frac{d\vec{r}}{dt}$$

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} \Rightarrow \frac{d}{dt}(-13 + 2t + 4t^2 - 3t^3)\hat{i}$$

$$= \frac{d}{dt}(2 + 8t - 9t^2)\hat{i} \quad \leftarrow \frac{d}{dt}(v)$$

$$\vec{a} = (8 - 18t)\hat{i}$$

$$\vec{F} = ma$$

$$= (0.15)(8 - 18t)\hat{i}$$

$$= (0.15)(8 - 18(3.4))\hat{i}$$

$$\boxed{\vec{F} = -7.98\hat{i}}$$

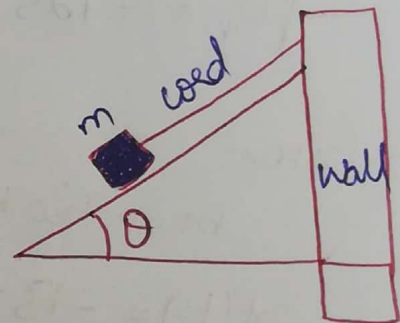
17- $m = 8.5 \text{ Kg}$

$$\theta = 30^\circ$$

a) $T = ?$

b) $F_N = ?$

c) If cord cut than resultant acceleration.



Sol

a) $T = mg \sin \theta$

$$= (8.5)(9.8) \sin 30^\circ$$

=

b) $F_N = mg \cos \theta \Rightarrow (8.5)(9.8) \cos 30^\circ \Rightarrow \boxed{72 \text{ N}}$

c) When cord cut no longer force exert and block accelerate ($T=0$).

$$-mg \sin \theta = ma$$

$$a = -g \sin \theta \Rightarrow -(9.8) \sin 30^\circ \Rightarrow \boxed{-4.9 \text{ m/s}^2}$$

given

$$m = 500 \text{ Kg}$$

$$V = 1600 \text{ Km/h} \Rightarrow 432 \text{ m/s}$$

$$t = 1.8 \text{ s}$$

$$|F_{\text{net}}| = ?$$

$$V = V_0 + at$$

↓
zero because it starts from rest.

Sol

$$V = at$$

$$a = \frac{V}{t}$$

$$F = ma$$

$$= m \left(\frac{V}{t} \right)$$

$$= 500 \left(\frac{432}{1.8} \right) \Rightarrow \boxed{1.2 \times 10^5 \text{ N}}$$

23-

$$\vec{W} = 820 \text{ N}$$

$$d = 20 \text{ m}$$

$$\theta = 22^\circ$$

$$T = 760 \text{ N}$$

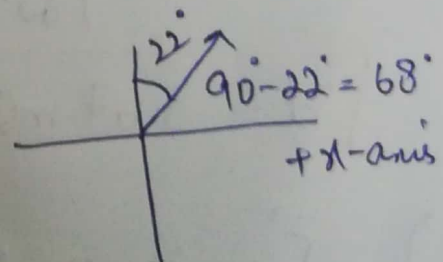
$$\vec{F} = ?$$

$$\vec{F}_{\text{net}} = ?$$

Sol

$$T = 760 \text{ N}$$

$$= T \cos \theta \hat{i} + T \sin \theta \hat{j}$$



$$= 760 \cos 68^\circ \hat{i} + 760 \sin 68^\circ \hat{j}$$

$$\boxed{\vec{T} = 285 \hat{i} + 705 \hat{j}}$$

b) The only force is due to Earth's gravity

$$\vec{F}_{\text{net}} = \vec{T} + \vec{W}$$

$$= 285 \hat{i} + 705 \hat{j} - 820 \hat{j}$$

$$\boxed{\vec{F}_{\text{net}} = 285 \hat{i} - 115 \hat{j}}$$

$$c) |\vec{F}_{\text{net}}| = 307 \text{ N}$$

$$d) \theta = \tan^{-1} \left(\frac{-115}{285} \right) \Rightarrow \boxed{-22^\circ}$$

$$e) \vec{a} = \vec{F}_{\text{net}} / m$$

$$W = mg$$

$$m = W/g \Rightarrow \boxed{83.7}$$

$$\text{So, } \vec{a} = 307 / 83.7$$

$$\boxed{\vec{a} = 3.67 \text{ ms}^{-2}}$$

f) $\vec{F}_{\text{net}}$ is parallel to $\vec{a}$ so angle is -22° .